Data Analytics - Case Study (BikeStores 2016-2018)

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Methodology

Top-down Approach

Step 1

Understanding the Business Process

Gaining a comprehensive understanding of the end-to-end process, starting from the initial purchase through to its finalization.

Step 2

Identify KPI Metrics

Systematically establish KPI metrics, determining precise indicators to measure both success and failure.

Step 3

Reviewing the data at a high-level overview

Conduct a comprehensive data analysis employing a top-down approach to systematically examine the information and extract meaningful insights.

Step 4

Identify Business Problems

Systematically identify and assess business challenges and concerns.

Step 5

Solutions & Recommendations

Several key insights that pave the way for strategic solutions and recommendations to enhance overall business performance.

BUSINESS PROCESS

- **Customers are engaging with our products, making selections from our available stocks**
- **Customer places an order for a bike**
- **Stores verification and processing the order such as assembling the bike.**
- **Order is either completed and shipped to the customer, or rejected if requirements aren't met (e.g., stock unavailable), shipment timing is tracked against the required date to monitor delivery punctuality**
- **Store periodically reviews inventory to identify slow-moving or unsold stock, informing restocking and clearance decisions**

KPI

Revenue Performance:

Gross Revenue, Net Revenue, Discount Value, Qty Sold

Sales:

Average Order Value, Discount rate, MoM/YoY Growth Rate

Product:

Sell-Through Rate, Dead stock value, Revenue contribution by model year

Overview - Topline Analysis

How is the performance of our Platform? (2016-2018)

Sales Performance

SKU Sold
1,615

Rev Before Disc
$8,347,189

Rev After Disc
$7,480,537

Avg. Amount/SKU
$5,312

Avg. order value
4,765

SKU Sell-Through Coverage
95%

Sell through rate
33,80%

Metrics

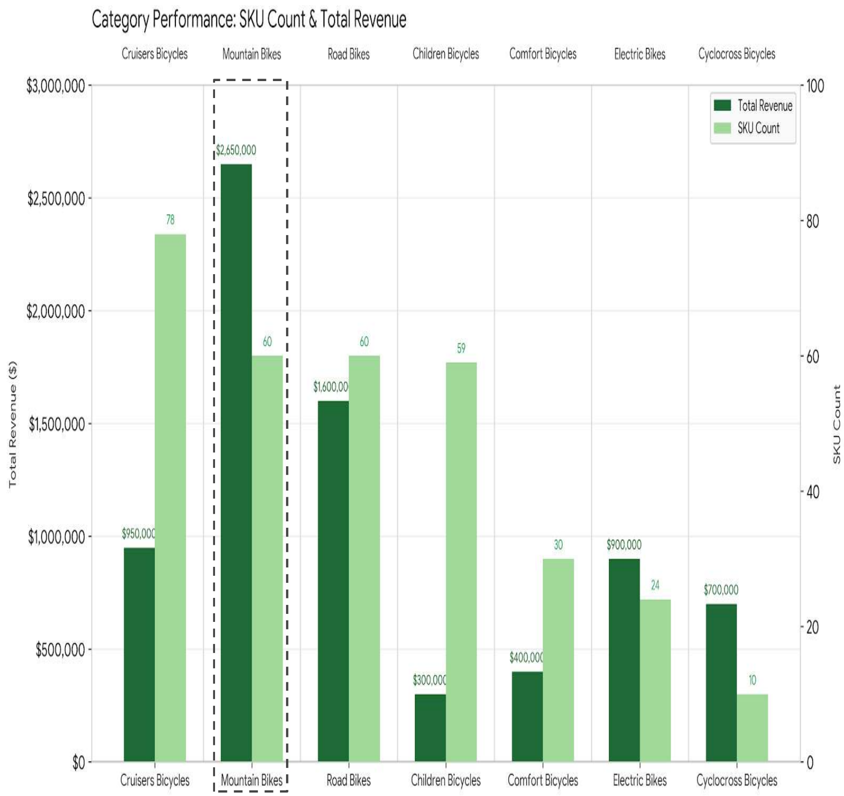
1. **SKU Sold**: Total number SKU sold
2. **Revenue Before Disc**: Price Before Disc
3. **Revenue After Disc**: Price After Disc (AVG Disc : 10%)
4. **Avg. Amount/SKU** : Total Price Before Disc / Total SKU Sold
5. **Avg.order value**: Revenue after disc / Total orders
6. **Sold-Out Rate**: Unique Item Sold / Unique Item Listed
7. **Sell Through Rate** : Qty sold / Qty sold + Total stock

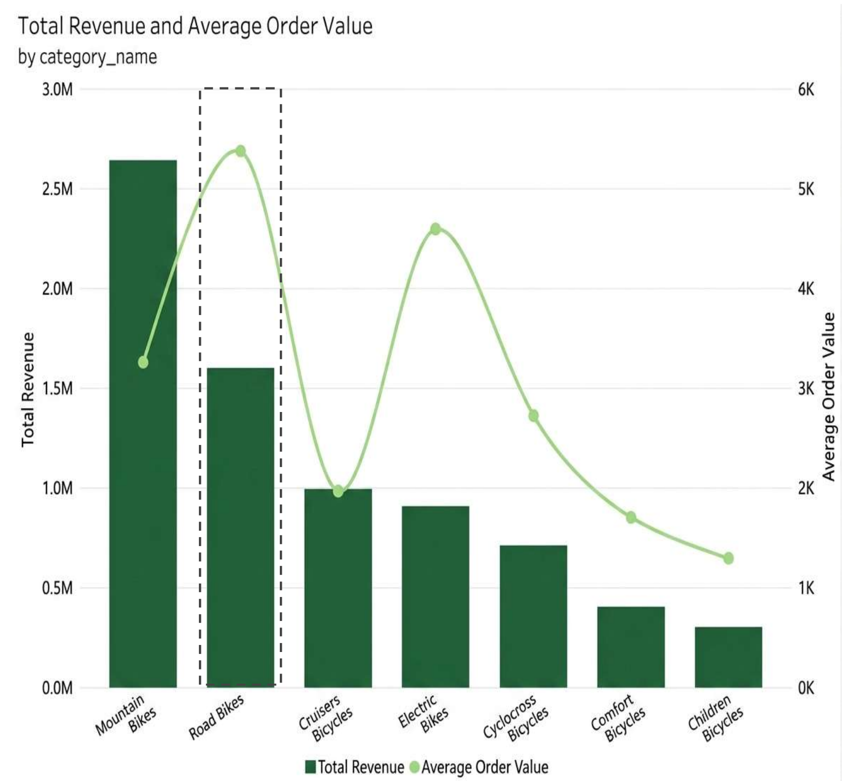
Key Takeaways

1. Our Platform sell around **1600 SKU,** about average **69 SKU / Month** contributing to more than **$267.000 revenue monthly.**
2. The SKU Sell-Through Coverage on our platform is comparatively **High**, standing at **95%.**
3. Our average order value around **4700+**, about average **56 total orders** monthly.
4. Sell-Through Rate is comparatively **Low**, standing at **33,80%.**

What are Our Customers Ordering?

24% of the order contribution is derived from the Mountain Bikes category



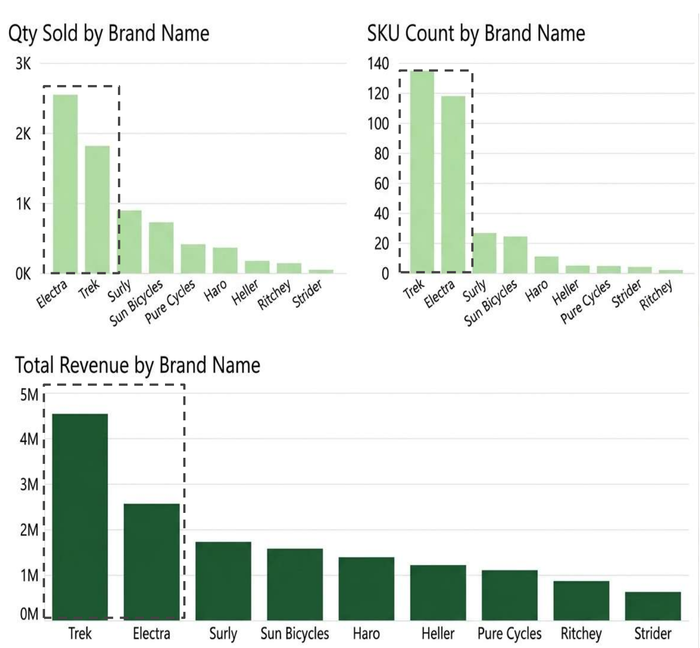


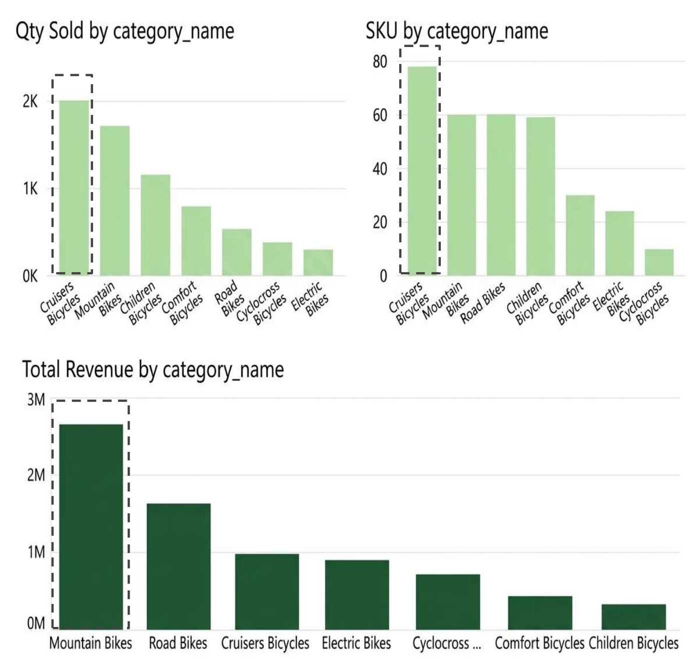
The Cruisers Bicycles category accounts for **24%** of total platform orders, while Mountain Bikes generate the highest revenue at approximately **$2.65 million**. This is primarily driven by the **higher price of Mountain Bike components.**

In the **Road Bikes** category, we observe the **highest average order value**, despite a relatively **lower order value.** This indicates road bikes has the highest average item price among our offerings and higher Qty per order among all categories.

Which Category are driving the highest Sales?

Top order contribution brands are mostly coming from Electra, Trek, and Surly





We have observed that the primary contributors to our **top-order brands** are **Electra and Trek**, with Electra leading in quantity sold at approximately **2,500 units**. Interestingly, **Trek** stands out as the **strongest revenue contributor**, generating approximately **$4.6 million**, despite having a lower quantity sold than Electra.

At the **category level**, **Cruisers Bicycles** records the highest quantity sold at around **2,000 units**, while **Mountain Bikes** generates the **highest revenue**, at approximately **$2.65 million**. This indicates that Mountain Bikes have a **higher average selling price**, allowing them to generate greater revenue despite lower unit sales.

Which market segment demonstrates the highest profitability?

60% of our profit came from Grocery Food Market

Topline	Total SKU Sold	% SKU Contribution	Qty Sold	Avg.Order Value	Total Revenue
Cruisers Bicycles	78	25,57%	2002	1042	$967,400
Mountain Bikes	57	18,69%	1714	3143	$2,652,342
Children Bicycles	54	17,70%	1152	459	$285,254
Road Bikes	53	17,38%	540	5286	$1,601,791
Comfort Bikes	30	9,84%	799	837	$386,731
Electric Bikes	23	7,54%	306	4575	$892,143
Cyclocross Bicycles	10	3,28%	385	2895	$694,877
Total	**305**	**100%**	**6898**	**4765**	**$7,480,537**

****Metrics:** Total Revenue = quantity * list_price * (1 – discount)

Our 2-year revenue stands at **$7 million**, translating to a revenue of **$24,526** per SKU sold. Notably, **Cruisers Bicycles** contributes nearly **26%** to our overall revenue. In contrast, **Cyclocross Bicycles** records the **lowest** contribution, comprisxing only **9.29%** of the total platform revenue.

Business Questions

1. Our sell-through rate averages around 34%, with some categories significantly lower, which product categories are underperforming, and what's driving Rp 11.4M in dead stock value?

2. Our discount rate sits at a flat ~10% across most products, is this discount strategy actually effective at driving sales volume, or is it simply eroding margin without a proportional return?

3. How is revenue and stock levels moving across model years, does the newest product line contribute the revenue ?

Which Product Categories are Underperforming?

Most products came from Road Bikes Category

A category is considered to be truly underperforming when it meets these three conditions

1. **Low level Sell-Through Rate,** shows a large proportion of unsold stocks
2. **Large total stock,** ensuring a low rate isn't just about having a small stock base
3. **Small revenue & Qty sold,** ensuring that this doesn't contribute much in terms of business value

****Metrics:** Dead Stock Value = unsold stock (low rate) * unit price of the product

category_name	Sell Through Rate	Total Stock	Total Revenue	Qty Sold
Road Bikes	20.52%	2091	1,601,791	540
Electric Bikes	21.64%	1108	892,143	306
Children Bicycles	30.64%	2608	285,254	1152
Cruisers Bicycles	37.21%	3378	967,400	2002
Comfort Bicycles	38.84%	1258	386,731	799
Mountain Bikes	39.24%	2654	2,652,342	1714
Cyclocross Bicycles	48.19%	414	694,877	385
Total	**33.80%**	**13511**	**7,480,537**	**6898**

category_name	Sum of Dead Stock Value
Road Bikes	4,137,073.45
Electric Bikes	2,506,692.46
Mountain Bikes	2,105,696.25
Cruisers Bicycles	1,360,793.27
Cyclocross Bicycles	453,285.85
Children Bicycles	432,255.10
Comfort Bicycles	409,174.32
Total	**11,404,970.70**

On the first table it shows that Road Bikes has the **lowest Sell through rate, considered high Total Stock,** and **highest Total Revenue**, this is a bit confusing, by rate it's the 'worst' but by revenue it's the 'highest'. A high revenue ≠ automatically not underperform. Road Bikes has the highest revenue because of **their high unit price**. 20,52% rate it's still the main cause, **almost 80% from 2091 stocks unit are still idle.** At this point, Road Bikes don't necessarily mean "total underperform".

But let's take a look at the second table

Road bikes are at the top far exceeding other categories – **4,131,073 out from 11,404,970** (nearly 36% from all dead stock value). On the first table, Road Bikes has 20.52% low rate, and generate the highest revenue, but their low sell-through rate and high dead stock value indicate inefficient inventory management. ($4.1M tied up in dead stock vs. only $1.6M in revenue)

Are discount strategies truly effective?

The store applied a flat discount strategy at 10-11%

- Our discount rate is at 10-11% across all category
- **The discount value is at $866,652**

The correlation analysis reveals a **neglible** relationship between the average discount rate and quantity sold. This implies that **increasing** the discount rate does not reliably drive higher quantities sold . Therefore, discounting in this dataset **does not appear to be an effective driver of sales volume**, suggesting the current discount policy may be reducing margin without generating a proportional increase in demand.

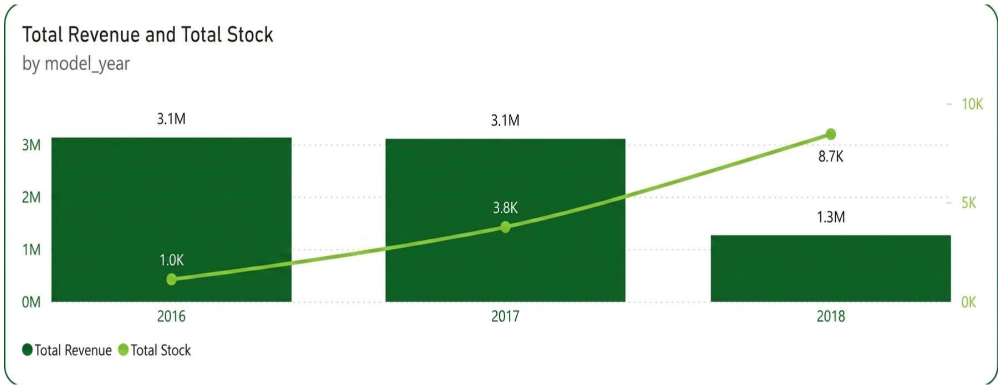
Category Name	**Avg.Disc Rate**	**Qty sold**	**Discount Value**
Children Bicycles	11%	1152	$34,814
Comfort Bicycles	10%	799	$43,854
Cruisers Bicycles	10%	2002	$110,826
Cycloross Bicycles	11%	385	$86,890
Electric Bikes	11%	306	$100,074
Mountain Bikes	11%	1714	$310,527
Road Bikes	11%	540	$179,665

The store truly applies a flat discount across all categories, because the rate is nearly same, the difference in discount value is **purely driven by the combination of product price and sales volume**. If we take a look, Electric Bikes have a same disc rate with Mountain Bikes, but the qty sold is much lower (306 vs 1,714). This is the further evidence that **discount don’t generate proportional volume**

if discounts were effective, categories with high discount values should consistently show high volumes as well.

How is revenue and stock levels moving across model years?

We're getting €0.09 less per order by giving discount, with assumption we sold the same number of SKU Items during both normal and discount periods



Model year	Total stock	Sell through rate	Total revenue
2016	1046	76,57%	$3,122,388
2017	3787	41,15%	$3,099,033
2018	8678	8,75%	$1,259,115

The line (stock) rose consistently from 2016-2018, while the bars (revenue) dropped sharply in 2018, opposite of the classic pattern which is "pilling up of old products". What's happening here is**, the latest model has the highest stock levels but the lowest revenue and lowest sell through rate**:

Few possibilities are:

- Overstocking for the new products – the store overbought but the market hasn't yet absorbed that much
- The 2018 data doesn't yet cover a full year

Recommendations, Hypotheses, & Validation Metrics

1. **Hypothesis:** The flat ~10% discount rate applied uniformly across categories is not an effective driver of sales volume, it may simply be eroding margin without a proportional return.
   Recommendation: Move from a uniform discount policy to a targeted discounting strategy , apply deeper discounts selectively to categories with high dead stock value (e.g., Road Bikes) to accelerate turnover, while reducing or eliminating discounts on categories with naturally strong demand (e.g., Cruisers Bicycles) where they add little incremental value.
   Validation Metric: Track Discount Rate % vs Qty Sold correlation per category post-implementation; monitor whether Net Revenue improves relative to Discount Value spent.

2. **Hypothesis:** Road Bikes carries the highest Dead Stock Value (Rp 4.13M, ~36% of total) due to a stock level disproportionate to actual demand, despite generating healthy absolute revenue.
   Recommendation: Reduce future procurement volume for Road Bikes and consider clearance pricing or bundling promotions on aging stock to free up capital tied in inventory. Further investigate whether this is a genuine demand mismatch or a seasonal/timing effect before making large procurement cuts.
   Validation Metric: Monitor Sell-Through Rate and Dead Stock Value for this category monthly; a successful correction should show Dead Stock Value trending down without a proportional drop in Total Revenue.

3. **Hypothesis:** The newest model year (2018) shows the highest stock and lowest revenue/sell-through rate, but this may be partially or fully explained by incomplete data coverage (a partial year) rather than genuine underperformance.
   Recommendation: Before taking inventory action on 2018 models, verify the actual date range of available order data. If data is incomplete, normalize the comparison using Avg Monthly Revenue instead of raw totals. Only after confirming a genuine gap should procurement or promotional decisions be made for newer models**.**
   Validation Metric: Avg Monthly Revenue and Sell-Through Rate by model year, recalculated once the full data range is confirmed; re-evaluate after data normalization.

Additional

1. Children Bicycles : High SKU Count, Low Revenue Contribution
   - **Hypothesis:** A large number of SKUs (60) in this category is not translating into proportional revenue, suggesting inefficient assortment, many low-performing variants diluting overall performance.
   - **Recommendation:** Conduct a SKU-level rationalization review, identify and discontinue the lowest sell-through variants in this category, and consolidate the assortment around better-performing products.
2. Cyclocross Bicycles : Misleading High Sell-Through Rate
   - **Hypothesis:** The category's high Sell-Through Rate (48.19%) is largely a function of its small stock base (414 units) rather than exceptionally strong demand — its absolute Qty Sold (385 units) is actually lower than several other categories.
   - **Recommendation:** Avoid over-interpreting this category as a "growth opportunity" based on rate alone. Any expansion decision should be validated against absolute demand signals (Qty Sold, Revenue trend) before increasing SKU count or stock investment.